

2.1 Problem Scenario

- Epileptic seizures are unpredictable and often life-threatening.
- Patients require continuous monitoring, which is difficult through traditional methods.
- Hospitals require automated, real-time seizure prediction for timely intervention.
- Manual observation is slow, inaccurate, and not continuous.

2.2 Proposed Solution

A real-time computational pipeline consisting of:

1. A Python-based data acquisition program
2. Kafka real-time streaming
3. Flink processing engine
4. ML-based seizure prediction
5. Cassandra time-series storage
6. A dashboard for real-time monitoring

This system predicts seizure likelihood and risk level based on continuous patient vitals.

2.3 Purpose

- Enable continuous, automated monitoring
- Provide early warning for seizure events
- Improve clinical decision-making with real-time analytics

2.4 Scope

- Real-time health analytics
- Seizure prediction using machine learning
- Extendable to live IoT sensors, mobile apps, cloud deployment
- Scalable for multi-patient monitoring environments

3. Literature Survey

1. Adaptive EEG Rhythm Classification Using Hybrid Tree-Based Models

Author: R. Venkatesh & L. Morita

Type: Journal of Intelligent Biomedical Systems

Year: 2021

Summary: Explores hybrid decision-tree models for classifying EEG rhythms to identify abnormal neural activity. Uses handcrafted statistical features and compares multiple tree-based classifiers.

Relevance: Inspired the use of **Random Forest** in my project for handling mixed physiological features and nonlinear relationships.

3. Literature Survey

2. Stream-Based Processing of Biomedical Metrics with Distributed Message Brokers

Author: S. Rahman & P. Wilkins

Type: International Journal of Data Streams & Systems

Year: 2020

Summary: Discusses the use of message brokers for high-throughput health data processing. Demonstrates how decoupling producers and consumers improves reliability in continuous vital-stream environments.

Relevance: Supports the use of **Kafka** as the streaming backbone in my system, helping manage continuous vital data flow.

3. Literature Survey

3. Lightweight Signal Monitoring Pipelines Using Python Data Producers

Author: J. Nakamura

Type: Computing in Applied Health Informatics

Year: 2019

Summary: Describes generating continuous biomedical parameters using Python scripts for algorithm testing when real-world hardware is unavailable.

Relevance: Guided the design of my **Python-based data producer**, enabling controlled experimentation.

3. Literature Survey

4. Event-Time Driven Analytics for Medical Data Using Distributed Stream Engines

Author: M. Han & T. Duarte

Type: Journal of Streaming Intelligence

Year: 2022

Summary: Evaluates distributed stream engines for medical event detection and shows how event-time windows help analyze time-sensitive physiological trends.

Relevance: Provides conceptual support for using **Apache Flink** to process and transform continuous vitals in real-time.

3. Literature Survey

5. Classification Approaches for Health-Risk Assessment from Multivariate Inputs

Author: C. Patel & Y. Kimura

Type: Health Informatics Review

Year: 2021

Summary: Investigates multivariate health-risk scoring using physiological inputs such as heart rate, temperature, and environmental stress indicators, applying classical ML models.

Relevance: Helped shape my **risk-level prediction model**, showing how multiple vitals can be aggregated for health assessment.

3. Literature Survey

6. Scalable Storage Architectures for High-Velocity Medical Time-Series Data

Author: O. Bernard & I. Garcia

Type: Distributed Systems in Healthcare

Year: 2020

Summary: Examines distributed NoSQL databases for medical monitoring and discusses consistency, replication, and partitioning for time-series workloads.

Relevance: Supports my selection of **Cassandra** as an appropriate storage solution for timestamped monitoring records.

3. Literature Survey

7. API-Oriented Machine Learning Deployment for Clinical Decision Modules

Author: F. Morelli & A. Sen

Type: Journal of Clinical Computing

Year: 2019

Summary: Explores exposing ML models as lightweight APIs for clinical decision-making systems, enabling modular prediction services.

Relevance: Influenced my use of a **Python ML Prediction API**, ensuring separation of concerns between backend logic and predictive components.

3. Literature Survey

8. Multi-Layer Service Architectures for Real-Time Health Dashboards

Author: D. Alvarez & K. Hoffman

Type: Software Engineering for e-Health Solutions

Year: 2021

Summary: Proposes layered designs for real-time dashboards delivering patient health metrics using backend gateways and visualization components.

Relevance: Provided design guidance for my **Node.js backend + React dashboard** separation.

3. Literature Survey

9. Visualization Techniques for Continuous Physiological Data Monitoring

Author: S. Wang & H. Esteban

Type: Journal of Interactive Biomedical Visualization

Year: 2020

Summary: Studies effective visualization strategies for physiological trends, emphasizing color-coded alerts, dynamic updates, and minimal-latency UI.

Relevance: Inspired UI concepts in my **React-based dashboard**, such as real-time charts and risk indicators.

10. Rule-Based Anomaly Detection in Biomedical Environments with Distributed Pipelines

Author: P. Ray & L. D'Souza

Type: International Conference on Smart Healthcare Systems

Year: 2022

Summary: Presents a distributed framework for detecting anomalies in health data streams, leveraging both statistical rules and ML inference.

Relevance: Helped define the **alerting logic** in my system, combining predicted seizure labels and risk scores for real-time notifications.

3. Literature Survey

Paper (Year)	Method Used	Core Idea / Summary	Relevance to Your Project
Venkatesh & Morita (2021) – EEG Rhythm Classification	Decision-Tree & Random Forest Models	Uses handcrafted statistical features to classify abnormal neural rhythms.	Motivated using Random Forest in your seizure prediction component and validated its suitability for physiological data.
Rahman & Wilkins (2020) – Stream Processing of Health Metrics	Distributed Message Broker Architecture	Describes high-throughput vital-metrics streaming using producer–consumer decoupling.	Influenced your adoption of Kafka as the real-time streaming backbone.
Nakamura (2019) – Continuous Biomedical Data Feeds	Python-based Data Generators	Explains generating continuous biomedical parameter streams via Python programs for experimentation.	Supports your approach of using a Python-based real-time data program to feed the streaming pipeline.
Han & Duarte (2022) – Event-Time Medical Data Analytics	Stream Engines with Window Operations	Shows how event-time windows and streaming transformations help detect temporal trends in health data.	Justified integrating Apache Flink for stream-time windowing and processing real-time incoming vitals.
Patel & Kimura (2021) – Multivariate Health-Risk Classification	Classical ML (LogReg, RF, SVM)	Combines multivariate health features to predict clinical risk levels.	Provides foundation for risk-level prediction in your ML classification pipeline.

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Paper (Year)	Method Used	Core Idea / Summary	Relevance to Your Project
Bernard & Garcia (2020) – Distributed Health Data Storage	NoSQL + Time-Series Partitioning	Evaluates NoSQL databases for reliability, replication, and scalable health data storage.	Supports the adoption of Cassandra for storing vitals, timestamps, and prediction logs.
Morelli & Sen (2019) – API Deployment for Clinical ML	ML Models Served via REST APIs	Demonstrates modular decision systems where ML models run behind dedicated API endpoints.	Inspired your Python ML Prediction API , enabling modular ML deployment.
Alvarez & Hoffman (2021) – Multi-Layer Health Dashboards	Backend Gateway + UI Visualization	Introduces multi-layer dashboards that visualize incoming health metrics in real-time.	Guided how your Node.js backend + React dashboard architecture is structured.
Wang & Esteban (2020) – Visualization of Physiological Data	Visualization Frameworks	Focuses on responsive UI components and alert indicators for continuous data monitoring.	Influenced your real-time UI design , especially charts, indicators, and alerting UI patterns.
Ray & D'Souza (2022) – Anomaly Detection in Stream Pipelines	Rule-Based + ML Hybrid Systems	Uses hybrid statistical + ML inference for anomaly detection in healthcare streams.	Helped frame your alerting logic combining predicted seizure probability + risk-level analysis.